

Education, Child Marriage, and Heterogeneous Wage Returns among Women in Nigeria

Table 1: Descriptive Statistics – Wage Earners

Variable	National (N=2,387)	N. Nigeria (N=932)
Continuous: Mean (SD)		
Log annual wage	12.02 (1.42)	11.83 (1.60)
Years of schooling	14.42 (5.81)	13.06 (7.02)
Age	36.69 (11.03)	36.34 (10.85)
Experience	16.01 (12.13)	16.65 (11.91)
Binary (%)		
Child marriage (%)	11.7	23.1
Rural (%)	49.9	56.3
Literate (%)	86.1	78.9
Currently married (%)	61.0	70.7
Religion (%)		
Religion: Islam (%)	76.5	53.1
Religion: Christianity (%)	23.0	46.8
Religion: Other (%)	0.5	0.1
Father education (%)		
Father edu: None (%)	33.6	38.1
Father edu: Primary (%)	43.6	28.2
Father edu: Secondary+ (%)	22.8	33.7
Mother education (%)		
Mother edu: None (%)	45.1	51.2
Mother edu: Primary (%)	40.9	28.2
Mother edu: Secondary+ (%)	14.0	20.6
Sector (%)		
Sector: Urban (%)	50.1	43.7
Sector: Rural (%)	49.9	56.3
N	2,387	932

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied.

Table 2: National – Quadratic Mincer

	Basic	+ Controls	+ State FE	+ Interaction	Heckman
Years of Schooling	0.116*** (0.006)	0.111*** (0.006)	0.089*** (0.007)	0.082*** (0.008)	0.077*** (0.003)
Years of Schooling ²	0.007*** (0.001)	0.007*** (0.001)	0.008*** (0.001)	0.009*** (0.001)	0.009*** (0.000)
Experience	0.042*** (0.002)	0.043*** (0.002)	0.040*** (0.002)	0.040*** (0.002)	0.020*** (0.001)
Experience ²	−0.001*** (0.000)	−0.001*** (0.000)	−0.001*** (0.000)	−0.001*** (0.000)	−0.001*** (0.000)
Child Marriage	−0.314*** (0.081)	−0.263*** (0.081)	−0.078 (0.084)	−0.144 (0.091)	
Years × Child Marriage				0.022* (0.011)	
Rural		−0.261*** (0.044)	−0.145*** (0.048)	−0.142*** (0.048)	
Inv. Mills Ratio ()					0.822** (0.361)
Currently Married		−0.098** (0.047)	−0.035 (0.046)	−0.035 (0.046)	
Observations	2387	2387	2387	2387	26 432
R ²	0.403	0.412	0.479	0.480	
Adj. R ²	0.402	0.411	0.470	0.470	
Residual Std. Error					1.222
State FE	No	No	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Interaction	No	No	No	Yes	No
Selection Correction	No	No	No	No	Yes

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 3: Northern Nigeria – Quadratic Mincer

	Basic	+ Controls	+ State FE	+ Interaction	Heckman
Years of Schooling	0.071*** (0.011)	0.066*** (0.011)	0.053*** (0.011)	0.043*** (0.013)	0.048*** (0.005)
Years of Schooling ²	0.012*** (0.001)	0.012*** (0.001)	0.011*** (0.001)	0.011*** (0.001)	0.009*** (0.001)
Experience	0.042*** (0.003)	0.041*** (0.003)	0.040*** (0.003)	0.039*** (0.003)	0.026*** (0.002)
Experience ²	−0.001*** (0.000)	−0.001*** (0.000)	−0.001*** (0.000)	−0.001*** (0.000)	−0.001*** (0.000)
Child Marriage	−0.262*** (0.094)	−0.237** (0.094)	−0.150 (0.096)	−0.236** (0.113)	
Years × Child Marriage				0.017 (0.012)	
Rural		−0.253*** (0.074)	−0.130* (0.073)	−0.123* (0.073)	
Inv. Mills Ratio ()					0.217 (0.420)
Currently Married		−0.187** (0.080)	−0.123 (0.078)	−0.113 (0.079)	
Observations	932	932	932	932	16 130
R ²	0.540	0.548	0.605	0.606	
Adj. R ²	0.537	0.544	0.594	0.594	
Residual Std. Error					1.021
State FE	No	No	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Interaction	No	No	No	Yes	No
Selection Correction	No	No	No	No	Yes

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 4: Average Marginal Effects on Pr(Wage Employment)

Variable	National			Northern Nigeria		
	AME (Nat.)	SE (Nat.)	p (Nat.)	AME (N.Nig.)	SE (N.Nig.)	p (N.Nig.)
Child Marriage	-0.0153	0.0048	0.001	-0.0080	0.0042	0.056
Experience	0.0025	0.0002	< 0.001	0.0023	0.0002	< 0.001
Experience Sq.	-0.0001	0.0000	< 0.001	-0.0001	0.0000	< 0.001
Rural	-0.0209	0.0040	< 0.001	-0.0099	0.0041	0.015
Years of Schooling	0.0098	0.0004	< 0.001	0.0041	0.0004	< 0.001
Years of Schooling Sq.	0.0011	0.0000	< 0.001	0.0008	0.0001	< 0.001

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01. AME via marginaleffects package.

Table 5: Generalized Random Forest: Average Conditional Association

Region	ATE	Std. Error	Return (%)
National	0.155	0.017	15.5%
Northern Nigeria	0.149	0.016	14.9%

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees. Source: NLSS 2018-19.

Variable Importance		
Variable	Importance (Nat.)	Importance (N.Nig.)
Experience	0.319	0.380
Mother's Education	0.230	0.161
Age	0.229	0.250
Father's Education	0.126	0.126
Child Marriage	0.064	0.046
Rural	0.032	0.037

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling. 4,000 trees. Source: NLSS 2018-19.

Table 6: Key Estimates – National vs. Northern Nigeria

Metric	National	N. Nigeria
Average Return (CF)	15.5%	14.9%
Return Std. Error	0.017	0.016
Child Marriage Penalty (OLS+FE)	-7.8%	-15.0%
Interaction: Years x Child Marriage	0.022	0.017
Interaction p-value	0.054	0.155
Child Marriage Var. Importance (CF)	6.4%	4.6%
Heckman lambda	0.822**	0.217 (n.s.)
N (wage earners)	2,387	932
N (selection sample)	26,450	16,140

Note:

Source: NLSS 2018-19. CF = Conditional-Association Forest.
 OLS+FE = OLS with state fixed effects. * $p < 0.1$, ** $p < 0.05$,
 *** $p < 0.01$.

Figure 1: Heterogeneous Returns to Education – National. Generalized random forest estimates by child marriage status, with 95% confidence bands. Source: NLSS 2018-19.

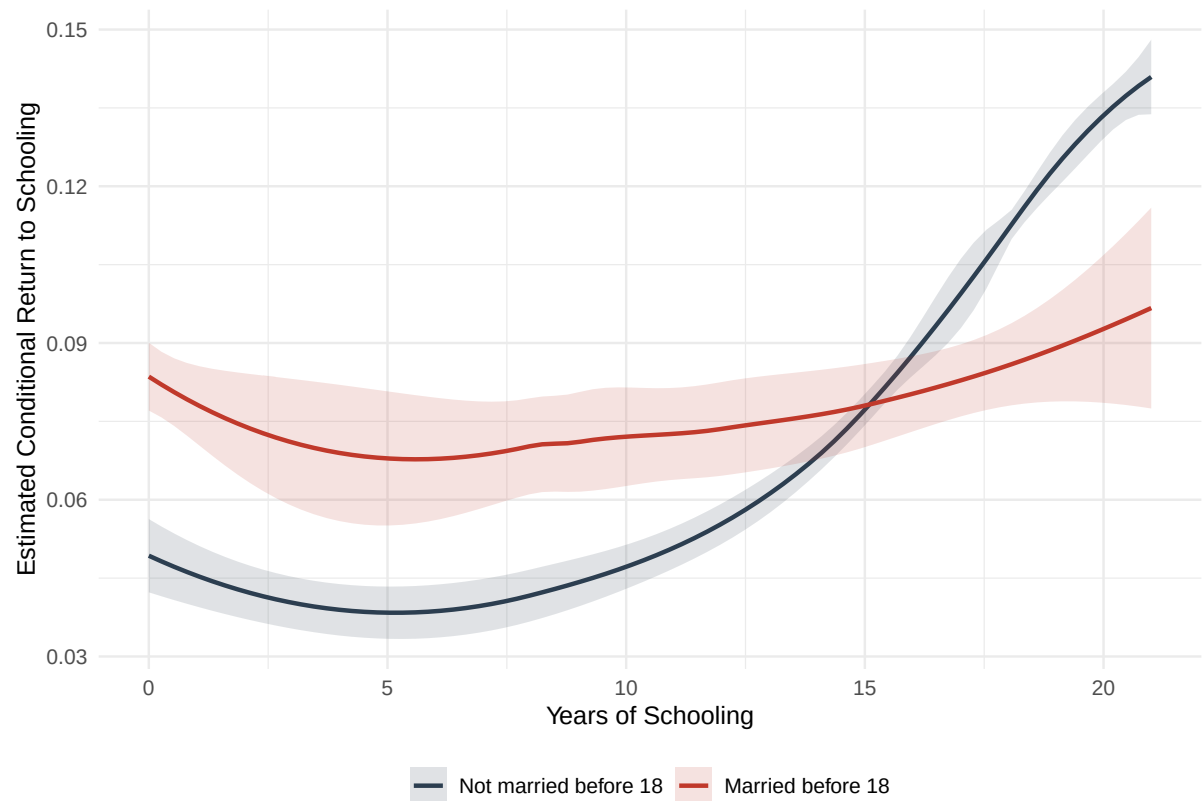


Figure 2: Heterogeneous Returns to Education – Northern Nigeria. Generalized random forest estimates by child marriage status, with 95% confidence bands. Source: NLSS 2018-19.

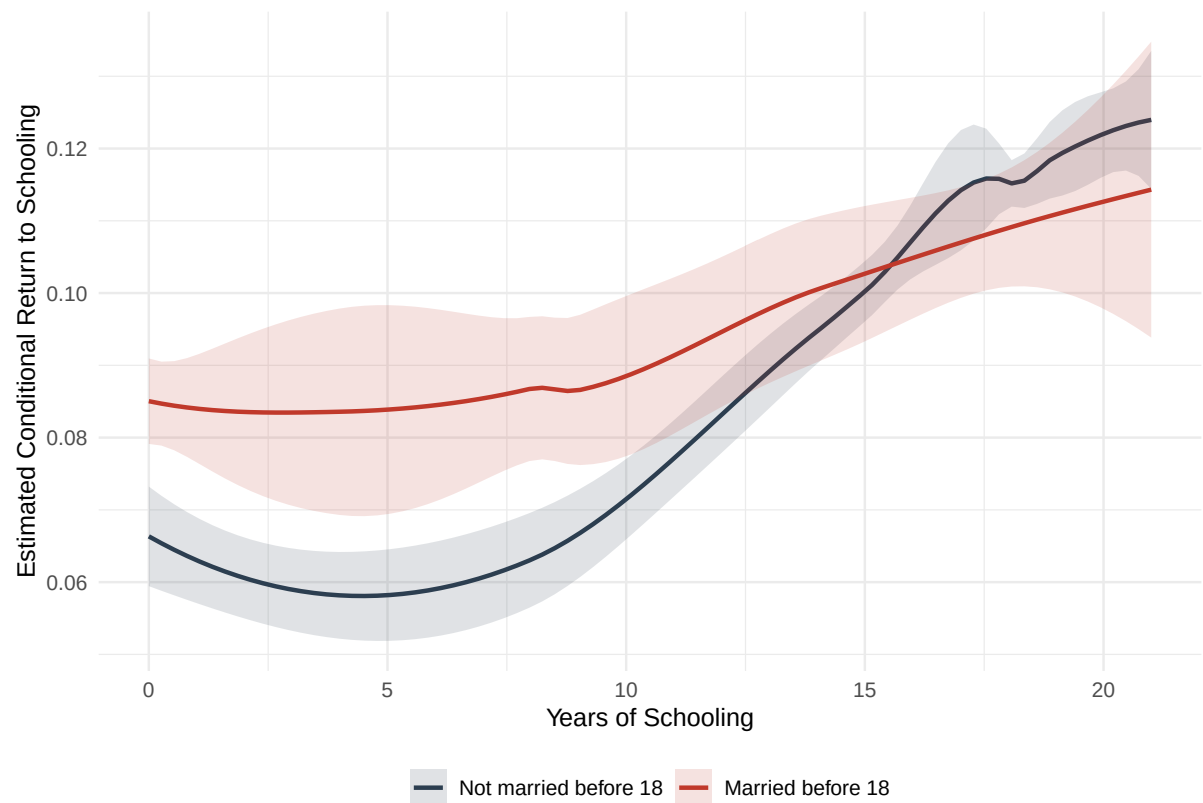


Figure 3: Returns Across Methods – National. Marginal returns from quadratic OLS and linear OLS, each with 95% confidence bands (delta method), plus a generalized random forest smooth (dashed, with band) for comparison. Blue = not married before 18; Red = married before 18. Source: NLSS 2018-19.

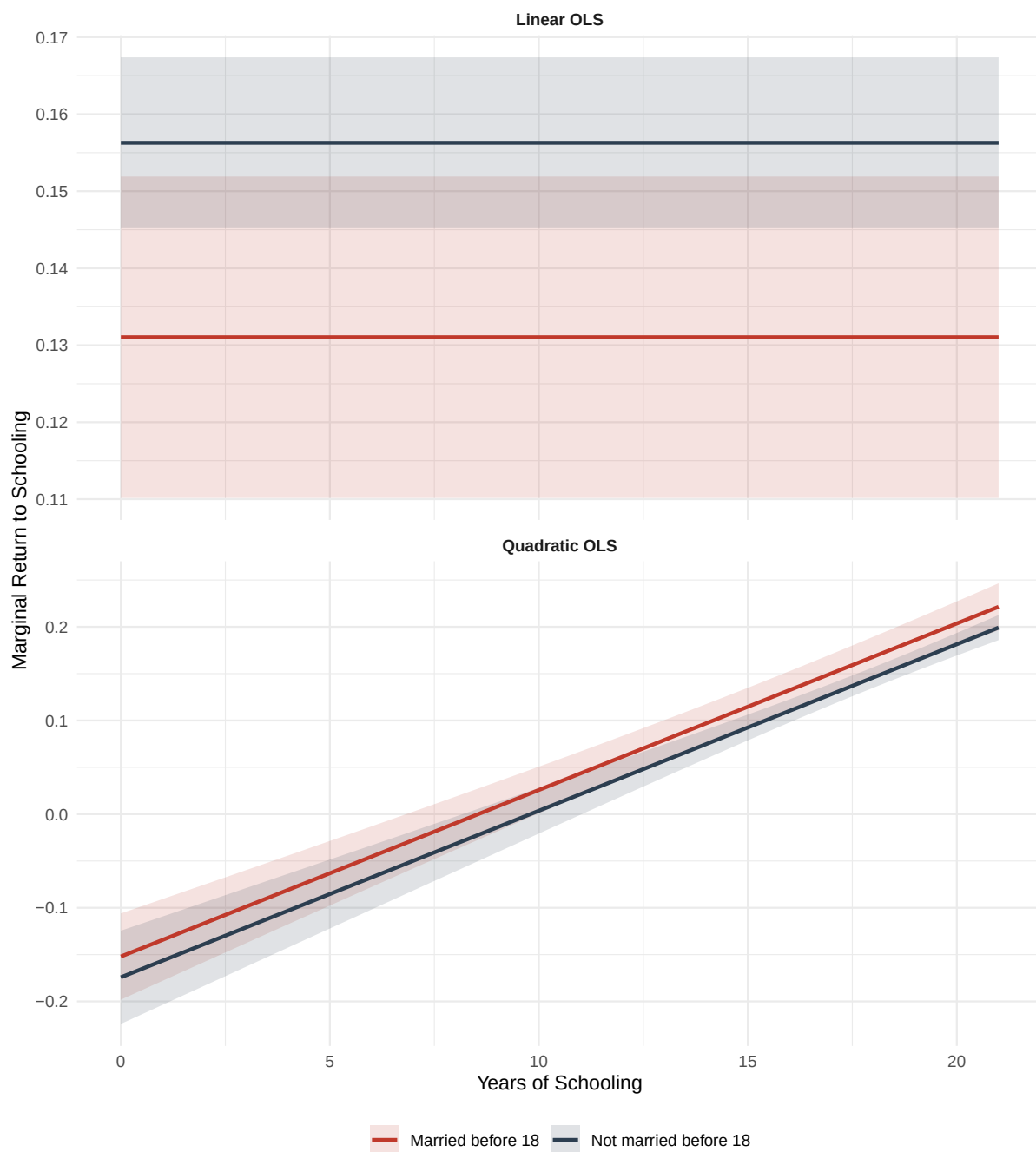
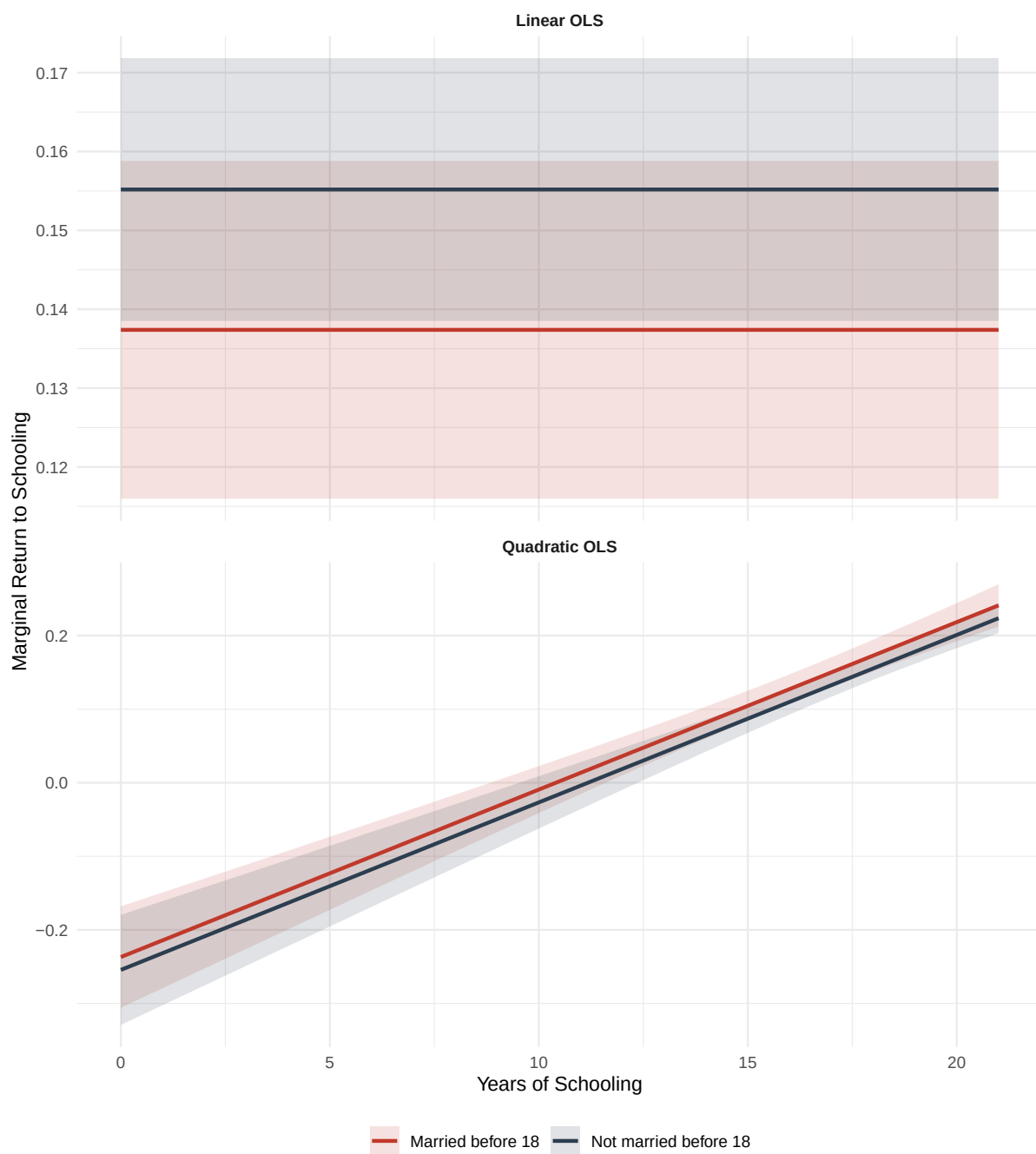


Figure 4: Returns Across Methods – Northern Nigeria. Marginal returns from quadratic OLS and linear OLS, each with 95% confidence bands (delta method), plus a generalized random forest smooth (dashed, with band) for comparison. Blue = not married before 18; Red = married before 18. Source: NLSS 2018-19.



Appendix: Additional Tables and Figures

Table A1: Sample Construction – National

Step	Description	N	N_dropped
0	Full Whole Country dataset	27,488	0
1	Women only	27,488	0
2	Age 18-64 (working age)	27,488	0
3	Non-missing key variables	26,450	1,038
4	Workers with valid wage data	2,387	24,063

Table A2: Sample Construction – Northern Nigeria

Step	Description	N	N_dropped
0	Full Northern Nigeria dataset	16,611	0
1	Women only	16,611	0
2	Age 18-64 (working age)	16,611	0
3	Non-missing key variables	16,140	471
4	Workers with valid wage data	932	15,208

Table A3: National – Linear Specification

	Basic	+ Controls	+ State FE	+ Interaction
Years of Schooling	0.169*** (0.005)	0.163*** (0.005)	0.152*** (0.005)	0.156*** (0.006)
Experience	0.047*** (0.002)	0.046*** (0.002)	0.044*** (0.002)	0.045*** (0.002)
Child Marriage	−0.120 (0.083)	−0.087 (0.083)	0.031 (0.087)	0.314** (0.154)
Years × Child Marriage				−0.025** (0.011)
Rural		−0.270*** (0.046)	−0.181*** (0.050)	−0.182*** (0.050)
State FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
Interaction	No	No	No	Yes
Currently Married		0.087* (0.046)	0.140*** (0.045)	0.135*** (0.046)
Observations	2387	2387	2387	2387
R ²	0.362	0.372	0.433	0.434
Adj. R ²	0.361	0.371	0.423	0.424

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Standard errors in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table A4: Northern Nigeria – Linear Specification

	Basic	+ Controls	+ State FE	+ Interaction
Years of Schooling	0.172*** (0.007)	0.166*** (0.007)	0.149*** (0.007)	0.155*** (0.009)
Experience	0.047*** (0.004)	0.046*** (0.004)	0.045*** (0.003)	0.045*** (0.003)
Child Marriage	−0.142 (0.102)	−0.112 (0.101)	−0.044 (0.103)	0.148 (0.170)
Years × Child Marriage				−0.018 (0.013)
Rural		−0.338*** (0.080)	−0.202** (0.079)	−0.207*** (0.079)
State FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
Interaction	No	No	No	Yes
Currently Married		0.041 (0.082)	0.060 (0.081)	0.050 (0.081)
Observations	932	932	932	932
R ²	0.459	0.470	0.539	0.540
Adj. R ²	0.457	0.467	0.527	0.527

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Returns by Education Level – National

	By Level
Primary	0.205 (0.216)
Secondary	0.309** (0.122)
Higher Ed	1.558*** (0.122)
Experience	0.041*** (0.002)
Experience Sq.	−0.001*** (0.000)
Child Marriage	−0.167* (0.093)
N	2387
R2	0.369

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Omitted: No education. State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Returns by Education Level – Northern Nigeria

	By Level
Primary	−0.319 (0.351)
Secondary	0.215 (0.150)
Higher Ed	1.912*** (0.139)
Experience	0.044*** (0.004)
Experience Sq.	−0.002*** (0.000)
Child Marriage	−0.190* (0.105)
N	932
R2	0.521

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Omitted: No education. State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Returns to Literacy – National

	Literacy
Literate	1.159*** (0.089)
Experience	0.029*** (0.002)
Experience Sq.	−0.001*** (0.000)
Child Marriage	−0.242** (0.095)
N	2387
R2	0.323

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Returns to Literacy – Northern Nigeria

	Literacy
Literate	1.529*** (0.119)
Experience	0.035*** (0.004)
Experience Sq.	−0.002*** (0.000)
Child Marriage	−0.309*** (0.110)
N	932
R2	0.461

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: State FE, rural, marital controls included. Source: NLSS 2018-19. Women aged 18-64 in wage employment. Survey weights applied. Schooling and experience are mean-centred. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Probit: Pr(Wage Employment) – National

	Pr(Wage Work)
Years of Schooling	0.076*** (0.003)
Years Sq.	0.009*** (0.000)
Experience	0.020*** (0.001)
Experience Sq.	−0.001*** (0.000)
Child Marriage	−0.123*** (0.040)
Rural	−0.158*** (0.029)
Observations	26 450
State FE	Yes
Parent Education	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Probit: Pr(Wage Employment) – Northern Nigeria

	Pr(Wage Work)
Years of Schooling	0.047*** (0.005)
Years Sq.	0.009*** (0.001)
Experience	0.026*** (0.002)
Experience Sq.	−0.001*** (0.000)
Child Marriage	−0.094* (0.050)
Rural	−0.111** (0.044)
Observations	16 140
State FE	Yes
Parent Education	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates. Probit includes schooling (+ sq.), experience (+ sq.), child marriage, urban/rural, parental education (factors), and state FE. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Generalized Random Forest: Calibration Test – National

Term	Estimate	Std. Error	t stat	p value
Mean Forest Prediction	1.045	0.184	5.678	0.00
Differential Forest Prediction	0.668	0.355	1.880	0.03

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling.
4,000 trees. Source: NLSS 2018-19.

Calibration Test – Northern Nigeria				
Term	Estimate	Std. Error	t stat	p value
Mean Forest Prediction	1.055	0.163	6.461	0.000
Differential Forest Prediction	0.969	0.432	2.243	0.013

Note: Outcome: Log Annual Wage. Treatment: Years of Schooling.
4,000 trees. Source: NLSS 2018-19.

Table A12: Descriptive Statistics – Full Working-Age Sample

Variable	National (N=26,450)	N. Nigeria (N=16,140)
Continuous: Mean (SD)		
Log annual wage	12.02 (1.42)	11.83 (1.60)
Years of schooling	8.37 (7.07)	5.90 (6.92)
Age	34.57 (12.26)	33.50 (11.69)
Experience	18.90 (14.26)	19.58 (13.69)
Binary (%)		
Child marriage (%)	31.6	46.2
Rural (%)	70.6	77.9
Literate (%)	59.7	47.5
Currently married (%)	69.3	78.4
Religion (%)		
Religion: Islam (%)	51.6	28.3
Religion: Christianity (%)	47.8	71.2
Religion: Other (%)	0.7	0.5
Father education (%)		
Father edu: None (%)	40.1	40.2
Father edu: Primary (%)	29.1	16.0
Father edu: Secondary+ (%)	30.8	43.8
Mother education (%)		
Mother edu: None (%)	50.6	51.3
Mother edu: Primary (%)	25.3	13.1
Mother edu: Secondary+ (%)	24.1	35.5
Sector (%)		
Sector: Urban (%)	29.4	22.1
Sector: Rural (%)	70.6	77.9
Labour market (%)		
Wage employment (%)	8.7	5.6
N	26,450	16,140

Note: Source: NLSS 2018-19. All women aged 18-64 with non-missing covariates (used in first-stage probit). Survey weights applied.

Figure A1: Distribution of Education Returns – National. Density of generalized random forest conditional associations by child marriage status. Source: NLSS 2018-19.

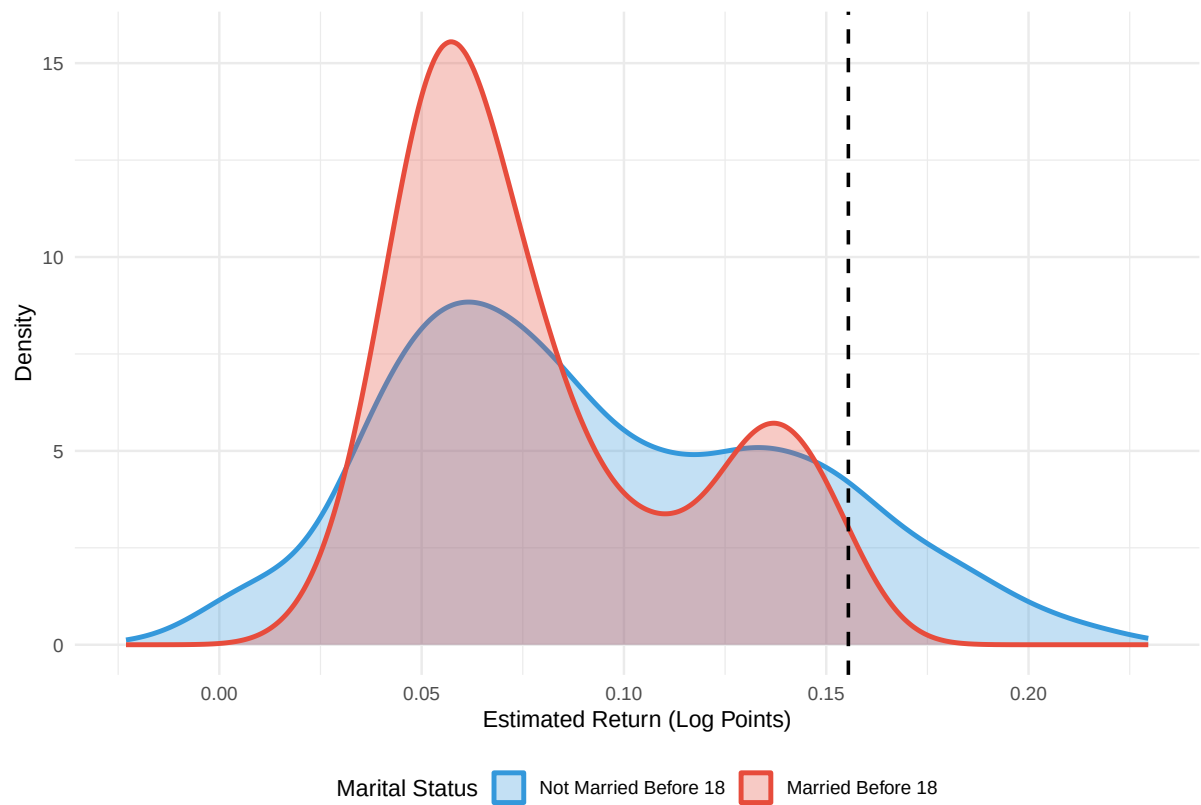


Figure A2: Returns by Level and Marriage Status – National. Average conditional associations (generalized random forest) by education level. Source: NLSS 2018-19.

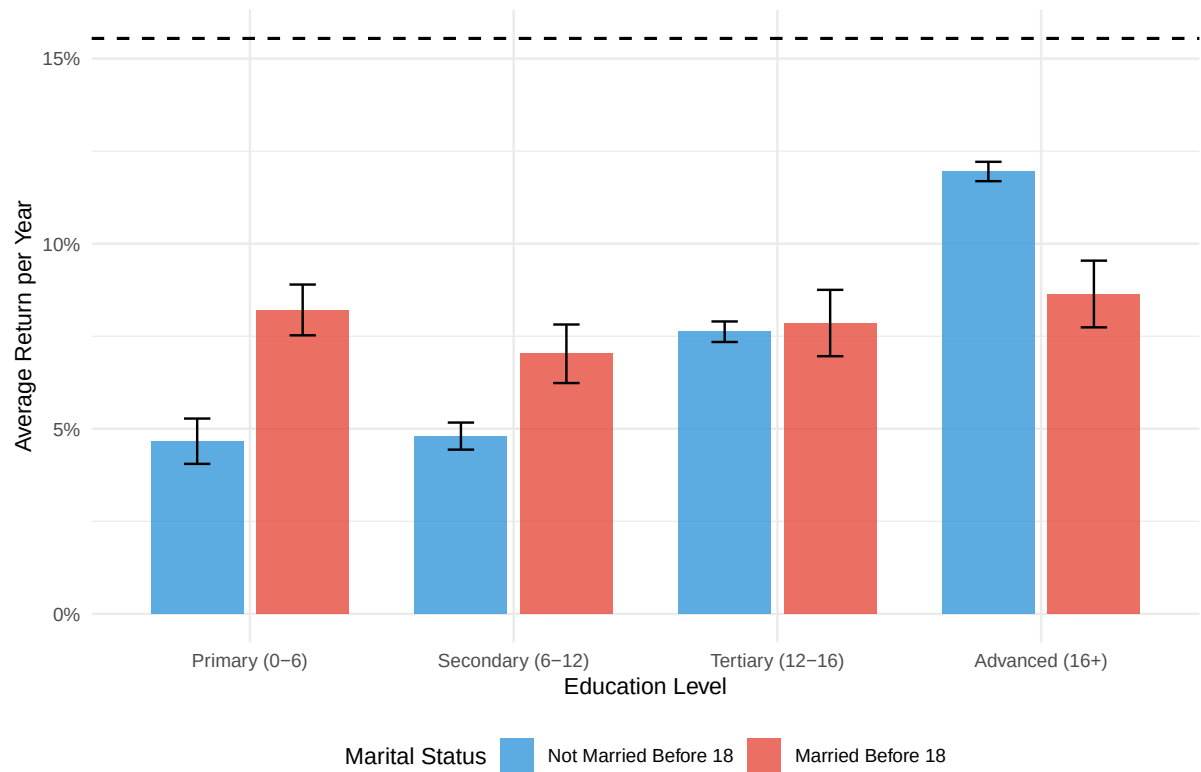


Figure A3: Predicted Returns vs. Wage Outcomes – National. Generalized random forest validation (conditional associations). Source: NLSS 2018-19.

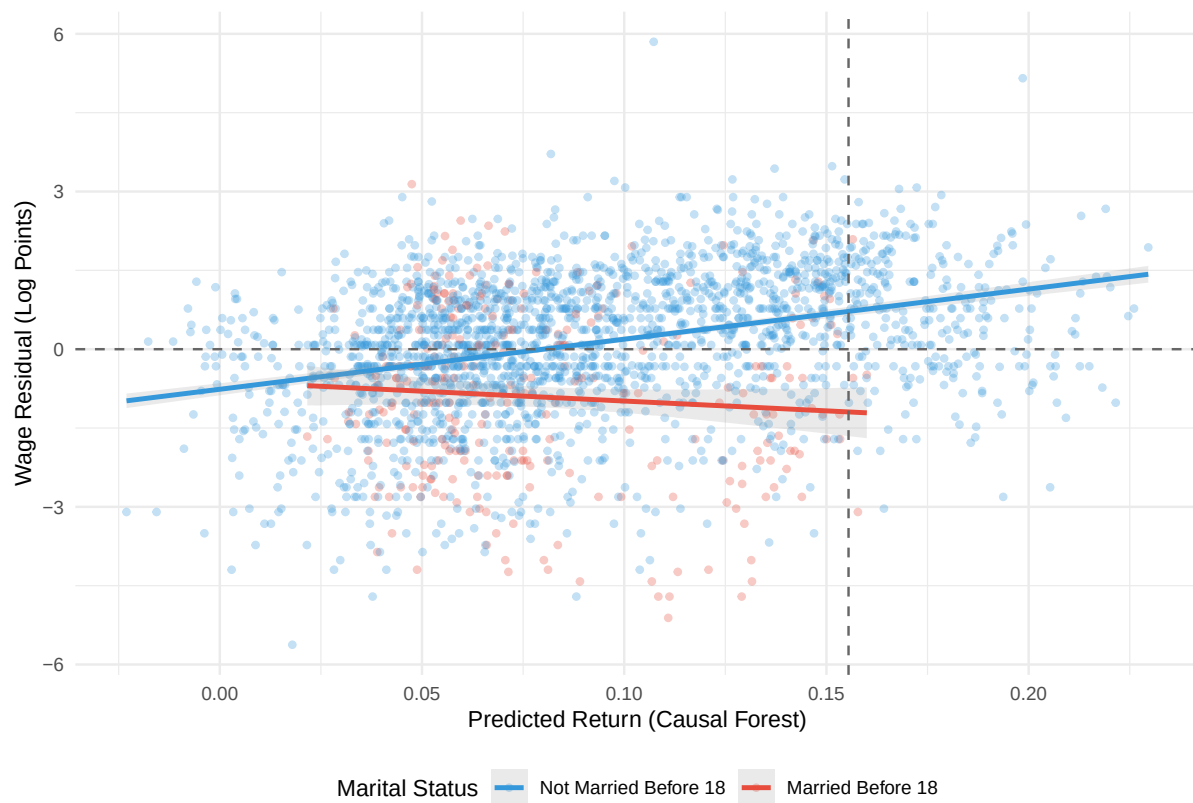


Figure A4: Distribution of Education Returns – Northern Nigeria. Density of generalized random forest conditional associations by child marriage status. Source: NLSS 2018-19.

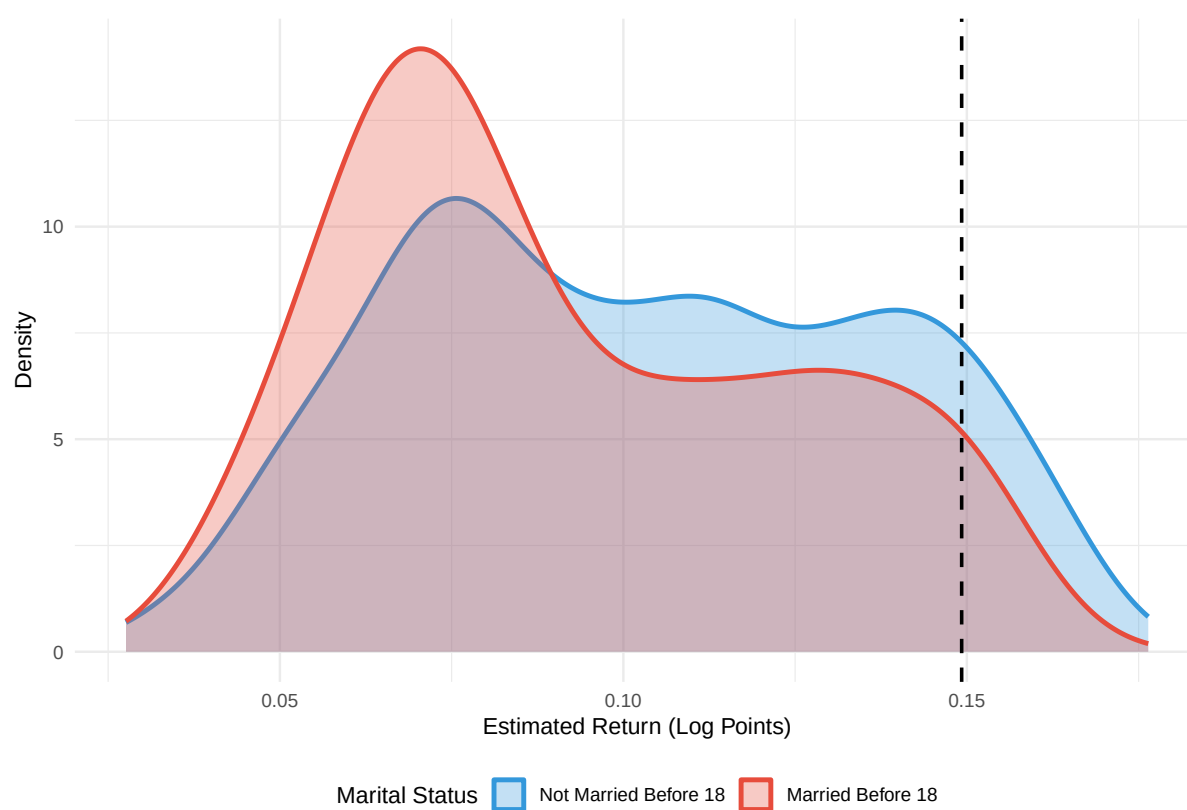


Figure A5: Returns by Level and Marriage Status – Northern Nigeria. Average conditional associations (generalized random forest) by education level. Source: NLSS 2018-19.

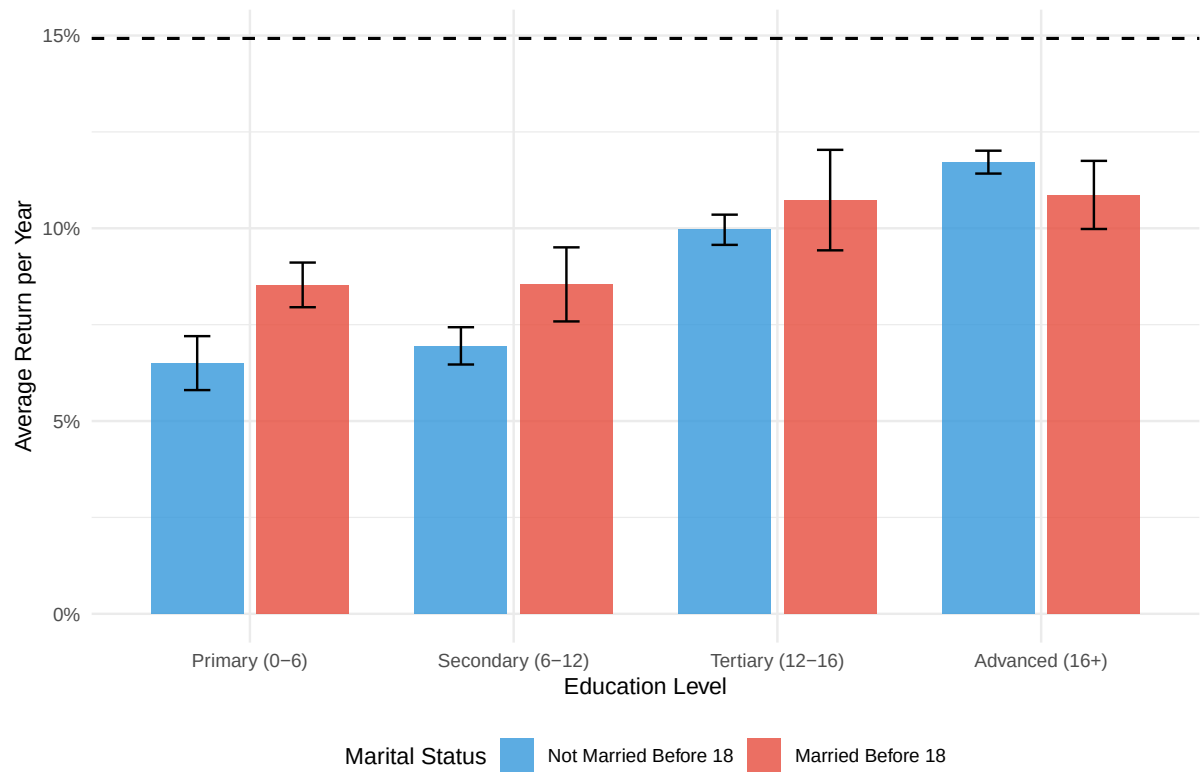


Figure A6: Predicted Returns vs. Wage Outcomes – Northern Nigeria. Generalized random forest validation (conditional associations). Source: NLSS 2018-19.

